Print guide

DIY CNC epoxy-granite mill · October 2026

Every printed part for the castings and the machine, cut to fit a Bambu A1 mini (180 mm cube; no piece is over 170 mm). The STL files are in `print/`, one folder per casting, with the list in `print/manifest.csv`. They are written by `model/build_cnc.py` on every run.

About 98 prints, roughly 3.6 kg of PETG. The filament figures are rough estimates.

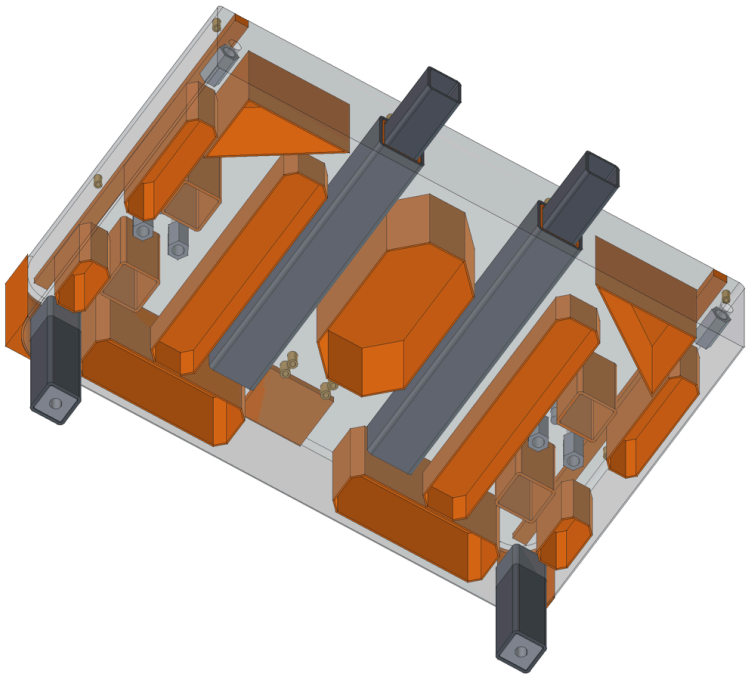
Settings

- **PETG for everything.** Thick sections of curing epoxy get warm, and PLA softens at about 55 °C.
- 0.4 mm nozzle, 0.2 mm layers.
- **Cores** (`core-*`): thin shells, 1.6 mm walls, so they print solid with any infill setting. They are already turned with the closed face on the bed.
- **Fillers, formers, plugs** (`filler-*`, `former-*`, `plug-*`): 3 walls, 15 % infill.
- **Liners and cast-in parts** (`liner-*`, `cast-in-*`): 4 walls, 40 % infill.
- Anything named `-partNofM` was cut to fit the bed. Glue the parts edge to edge with CA or epoxy and tape over the seam. Check the joined piece against the drawing before casting.

How the pieces are used

- The mould boxes are 18 mm melamine board, not printed: the flat board is what the important faces are cast against. Box sizes (inside) and casting orientation are in the table below.
- **Removable pieces** (formers, fillers, plugs): screw or hot-glue them to the board, then 3-5 coats of paste wax and PVA like the rest of the mould. Reuse the fillers for the second upright.
- **Pieces that stay in** (cores, liners, cast-in parts): no wax; scuff and degrease them so the epoxy grips. Cores float: fill them with dry sand and hold them down with a bridge across the mould.

Casting	Mould box inside (mm)	Cast with
Base, front half	600 x 345 x 110	top face down
Base, rear half	600 x 380 x 110	top face down
Uprights (both the same: print quantities are for two)	80 x 285 x 246	foot down
Beam	530 x 140 x 140	front (rail) face down



Foam cores instead of printed ones

The pocket cores are about 1.5 kg of the PETG. The rectangular ones are quick to cut from XPS foam board on a table saw or with a hot wire; wrap them in packing tape and wax them. The corners are chamfered 8-10 mm (30 mm on the rear centre core), and the triangles are right triangles. Glue foam cores to the bridge so they cannot float. Sizes (outline x depth from the underside):

Casting	Core	Outline (mm)	Depth (mm)
base-front	centre	100 x 260	85
base-front	outboard-L	130 x 260	85
base-front	trough-band-L	101 x 55	40
base-front	draw-rod-L	37 x 255	40
base-front	outboard-R	130 x 260	85
base-front	trough-band-R	101 x 55	40
base-front	draw-rod-R	37 x 255	40
base-rear	centre	100 x 185	85
base-rear	triangle-L	115 x 80	85
base-rear	knee-L	50 x 258	40
base-rear	outer-a-L	35 x 128	65
base-rear	outer-b-L	35 x 60	65
base-rear	rear-L	170 x 52	85

Casting	Core	Outline (mm)	Depth (mm)
base-rear	triangle-R	115 x 80	85
base-rear	knee-R	50 x 258	40
base-rear	outer-a-R	35 x 128	65
base-rear	outer-b-R	35 x 60	65
base-rear	rear-R	170 x 52	85

Base, front half

Folder `print/base-front/`, about 1245 g.

File	What	Qty	Size (mm)	Notes
`core-centre-part1of2.stl`	Pocket core (shell, stays in)	1	100 x 130 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-centre-part2of2.stl`	Pocket core (shell, stays in)	1	100 x 130 x 85	
`core-outboard-L-part1of2.stl`	Pocket core (shell, stays in)	1	130 x 130 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-outboard-L-part2of2.stl`	Pocket core (shell, stays in)	1	130 x 130 x 85	
`core-trough-band-L.stl`	Pocket core (shell, stays in)	1	101 x 55 x 40	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-draw-rod-L-part1of2.stl`	Pocket core (shell, stays in)	1	37 x 128 x 40	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-draw-rod-L-part2of2.stl`	Pocket core (shell, stays in)	1	37 x 128 x 40	
`core-outboard-R-part1of2.stl`	Pocket core (shell, stays in)	1	130 x 130 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-outboard-R-part2of2.stl`	Pocket core (shell, stays in)	1	130 x 130 x 85	
`core-trough-band-R.stl`	Pocket core (shell, stays in)	1	101 x 55 x 40	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-draw-rod-R-part1of2.stl`	Pocket core (shell, stays in)	1	37 x 128 x 40	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-draw-rod-R-part2of2.stl`	Pocket core (shell, stays in)	1	37 x 128 x 40	

File	What	Qty	Size (mm)	Notes
`former-gutter-L-part1of2.stl`	Former (removable)	1	15 x 162 x 15	Stick to the mould floor (the base is cast top-down), wax.
`former-gutter-L-part2of2.stl`	Former (removable)	1	15 x 162 x 13	
`former-gutter-R-part1of2.stl`	Former (removable)	1	15 x 162 x 15	Stick to the mould floor (the base is cast top-down), wax.
`former-gutter-R-part2of2.stl`	Former (removable)	1	15 x 162 x 13	
`former-trough-part1of4.stl`	Former (removable)	1	140 x 35 x 20	Stick to the mould floor (the base is cast top-down), wax.
`former-trough-part2of4.stl`	Former (removable)	1	140 x 35 x 25	
`former-trough-part3of4.stl`	Former (removable)	1	140 x 35 x 25	
`former-trough-part4of4.stl`	Former (removable)	1	140 x 35 x 22	
`filler-corner-1.stl`	Corner filler (removable)	1	40 x 40 x 110	Rounded corner of the mould box; wax.
`filler-corner-2.stl`	Corner filler (removable)	1	40 x 40 x 110	Rounded corner of the mould box; wax.
`liner-drain.stl`	Liner (stays in)	1	14 x 14 x 73	Drain liner; sits on the nipple and stays in.
`former-split-gasket-groove-part1of4.stl`	Former (removable)	1	145 x 2 x 14	Stick to the split-face mould wall, wax.
`former-split-gasket-groove-part2of4.stl`	Former (removable)	1	145 x 2 x 4	
`former-split-gasket-groove-part3of4.stl`	Former (removable)	1	145 x 2 x 4	
`former-split-gasket-groove-part4of4.stl`	Former (removable)	1	145 x 2 x 14	
`cast-in-base-front-tube-plug-l.stl`	Cast-in part (stays in)	2	42 x 10 x 42	Cast in, stays.
`cast-in-base-front-spigot-sleeve-l.stl`	Cast-in part (stays in)	2	44 x 69 x 44	Cast in, stays.
`cast-in-leg-front-l-plug.stl`	Cast-in part (stays in)	2	30 x 30 x 10	Cast in, stays.

Base, rear half

Folder `print/base-rear/`, about 1222 g.

File	What	Qty	Size (mm)	Notes
`core-centre-part1of2.stl`	Pocket core (shell, stays in)	1	100 x 92 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-centre-part2of2.stl`	Pocket core (shell, stays in)	1	100 x 92 x 85	
`core-triangle-L.stl`	Pocket core (shell, stays in)	1	115 x 80 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-knee-L-part1of2.stl`	Pocket core (shell, stays in)	1	50 x 129 x 40	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-knee-L-part2of2.stl`	Pocket core (shell, stays in)	1	50 x 129 x 40	
`core-outer-a-L.stl`	Pocket core (shell, stays in)	1	35 x 128 x 65	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-outer-b-L.stl`	Pocket core (shell, stays in)	1	35 x 60 x 65	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-rear-L.stl`	Pocket core (shell, stays in)	1	170 x 52 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-triangle-R.stl`	Pocket core (shell, stays in)	1	115 x 80 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-knee-R-part1of2.stl`	Pocket core (shell, stays in)	1	50 x 129 x 40	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-knee-R-part2of2.stl`	Pocket core (shell, stays in)	1	50 x 129 x 40	
`core-outer-a-R.stl`	Pocket core (shell, stays in)	1	35 x 128 x 65	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-outer-b-R.stl`	Pocket core (shell, stays in)	1	35 x 60 x 65	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`core-rear-R.stl`	Pocket core (shell, stays in)	1	170 x 52 x 85	Shell core, stays in the casting. Fill with dry sand while casting so it cannot float.
`former-gutter-L-part1of3.stl`	Former (removable)	1	15 x 120 x 12	Stick to the mould floor (the base is cast top-down), wax.
`former-gutter-L-part2of3.stl`	Former (removable)	1	15 x 120 x 10	
`former-gutter-L-part3of3.stl`	Former (removable)	1	15 x 120 x 9	
`former-gutter-R-part1of3.stl`	Former (removable)	1	15 x 120 x 12	Stick to the mould floor (the base is cast top-down), wax.
`former-gutter-R-part2of3.stl`	Former (removable)	1	15 x 120 x 10	
`former-gutter-R-part3of3.stl`	Former (removable)	1	15 x 120 x 9	
`former-motor-relief.stl`	Former (removable)	1	72 x 60 x 8	Stick to the mould floor (the base is cast top-down), wax.

File	What	Qty	Size (mm)	Notes
`filler-corner-1.stl`	Corner filler (removable)	1	40 x 40 x 110	Rounded corner of the mould box; wax.
`filler-corner-2.stl`	Corner filler (removable)	1	40 x 40 x 110	Rounded corner of the mould box; wax.
`cast-in-base-rear-tube-plug-l.stl`	Cast-in part (stays in)	2	42 x 10 x 42	Cast in, stays.
`cast-in-base-rear-spigot-sleeve-l.stl`	Cast-in part (stays in)	2	44 x 69 x 44	Cast in, stays.
`cast-in-leg-rear-l-plug.stl`	Cast-in part (stays in)	2	30 x 30 x 10	Cast in, stays.
`cast-in-socket-liner-l1.stl`	Cast-in part (stays in)	4	44 x 44 x 80	Cast in, stays.

Uprights (both the same: print quantities are for two)

Folder `print/upright/`, about 592 g.

File	What	Qty	Size (mm)	Notes
`filler-front-flare.stl`	Corner filler (removable)	2	80 x 70 x 70	Forms the curved flare at the front of the foot; wax. Above it, fill the box corner with a 70 x 168 x 80 mm wood block.
`filler-back-slope-part1of2.stl`	Corner filler (removable)	2	80 x 18 x 123	Forms the sloped back. Optional: a melamine board set at the same angle does the same job.
`filler-back-slope-part2of2.stl`	Corner filler (removable)	2	80 x 35 x 123	
`former-foot-gasket-groove-part1of2.stl`	Former (removable)	2	68 x 136 x 2	Stick to the mould floor (the upright is cast foot-down), wax.
`former-foot-gasket-groove-part2of2.stl`	Former (removable)	2	68 x 136 x 2	
`plug-tie-rod-pocket.stl`	Pocket plug (removable)	4	32 x 32 x 18	Forms the nut pocket; slides down the greased tie-rod core rod. Wax; pull after curing.
`cast-in-post-l1-plug.stl`	Cast-in part (stays in)	4	30 x 30 x 10	Cast in, stays.

Beam

Folder `print/beam/`, about 308 g.

File	What	Qty	Size (mm)	Notes
`filler-back-edge-1-part1of4.stl`	Corner filler (removable)	1	132 x 40 x 40	Rounded back edge of the beam mould; wax.
`filler-back-edge-1-part2of4.stl`	Corner filler (removable)	1	132 x 40 x 40	
`filler-back-edge-1-part3of4.stl`	Corner filler (removable)	1	132 x 40 x 40	
`filler-back-edge-1-part4of4.stl`	Corner filler (removable)	1	132 x 40 x 40	
`filler-back-edge-2-part1of4.stl`	Corner filler (removable)	1	132 x 40 x 40	Rounded back edge of the beam mould; wax.
`filler-back-edge-2-part2of4.stl`	Corner filler (removable)	1	132 x 40 x 40	
`filler-back-edge-2-part3of4.stl`	Corner filler (removable)	1	132 x 40 x 40	
`filler-back-edge-2-part4of4.stl`	Corner filler (removable)	1	132 x 40 x 40	

Machine parts

Folder `print/machine/`, about 184 g.

File	What	Qty	Size (mm)	Notes
`part-chain-mount-fixed.stl`	Machine part	1	25 x 38 x 21	
`part-chain-mount-moving.stl`	Machine part	1	37 x 123 x 45	
`part-sump-baffle-part1of2.stl`	Machine part	1	148 x 4 x 73	
`part-sump-baffle-part2of2.stl`	Machine part	1	148 x 4 x 73	
`part-sump-filter-frame-part1of2.stl`	Machine part	1	148 x 6 x 92	
`part-sump-filter-frame-part2of2.stl`	Machine part	1	148 x 6 x 92	